Plataformas de Desenvolvimento

24/25

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Mestrado em Informática e Sistemas

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Docker Swarm

- Why Swarm?
- Docker Swarm
- Docker Service
- Docker Stack?
- Importance for CI/CD
- Swarm Deprecated?

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Why Docker Swarm?

- How do we manage container lifecycle?
- How do we scale easily number of containers?
  - In several machines (up/down)
  - Inside the machine (up/down)
- How can we define where the containers get started (VMs, nodes, etc)?
- How do we achieve Zero Downtime Deployments?

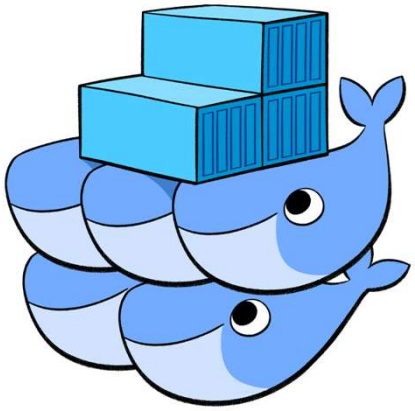
Docker Swarm

*A swarm consists of multiple Docker hosts which run in **swarm mode** and act as managers (to manage membership and delegation) and workers (which run swarm services). A given Docker host can be a manager, a worker, or perform both roles.*

Built-in orchestration

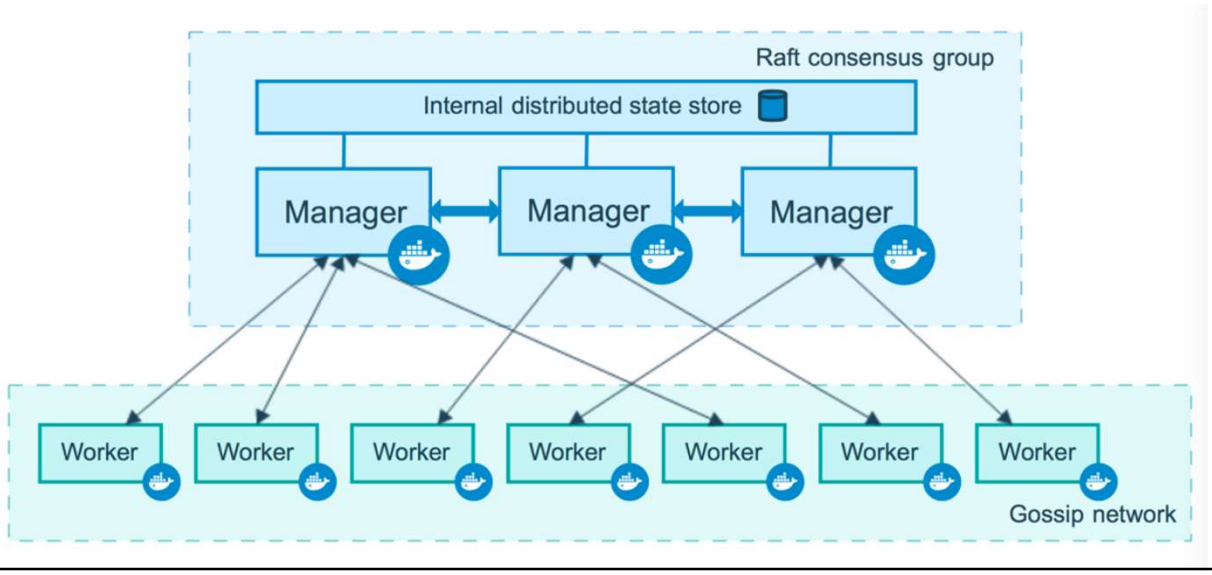
New commands available (not by default):

- docker swarm
- docker node
- docker service
- docker stack
- docker secret



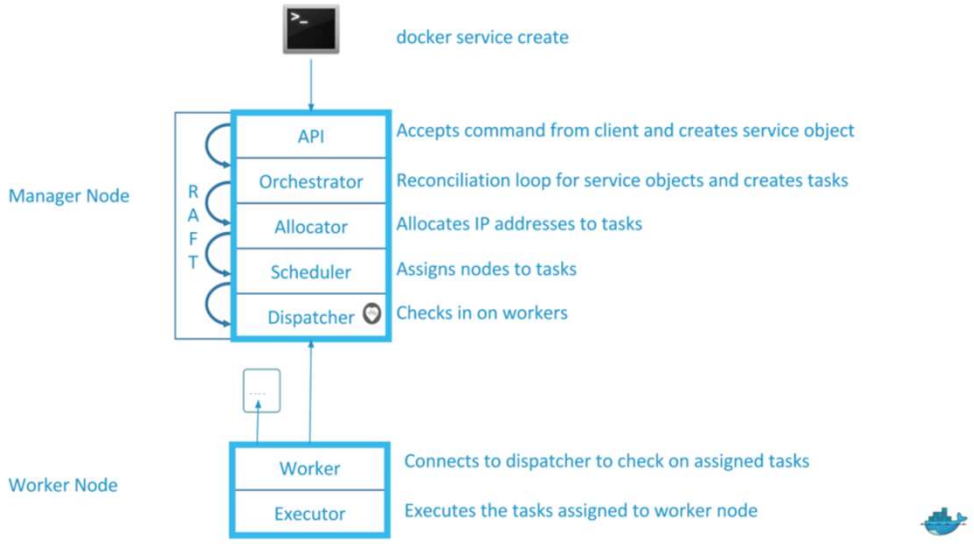
https://docs.docker.com/engine/swarm/

Docker Swarm



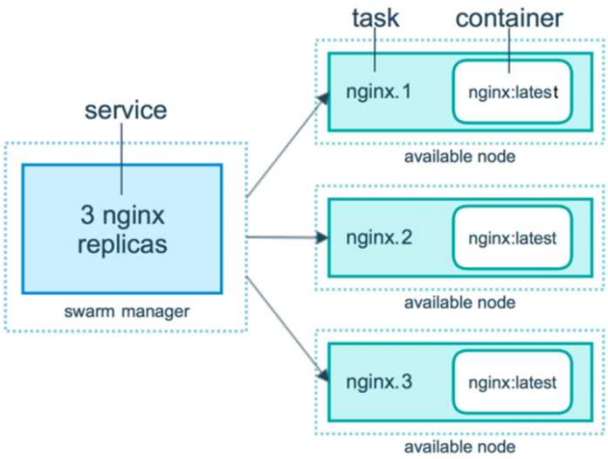
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Docker Swarm



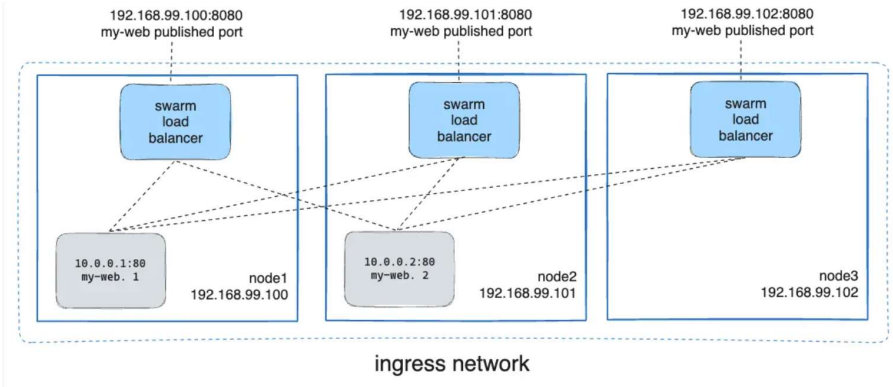
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Docker Service



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Docker Service



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Docker Service

Services and tasks

A **service** is the definition of the tasks to execute on the manager or worker nodes. It is the central structure of the swarm system and the primary root of user interaction with the swarm.

When you create a service, you specify which container image to use and which commands to execute inside running containers.

In the **replicated services** model, the swarm manager distributes a specific number of replica tasks among the nodes based upon the scale you set in the desired state.

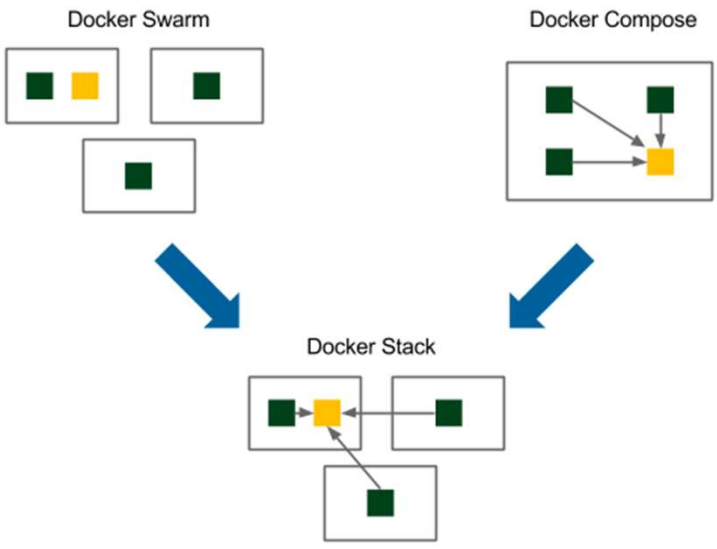
For **global services**, the swarm runs one task for the service on every available node in the cluster.

A **task** carries a Docker container and the commands to run inside the container. It is the atomic scheduling unit of swarm. Manager nodes assign tasks to worker nodes according to the number of replicas set in the service scale. Once a task is assigned to a node, it cannot move to another node. It can only run on the assigned node or fail.

https://docs.docker.com/engine/swarm/key-concepts/#services-and-tasks

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Docker Stack



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Docker Stack

Deploy a stack to a swarm

Estimated reading time: 7 minutes

When running Docker Engine in swarm mode, you can use `docker stack deploy` to deploy a complete application stack to the swarm. The `deploy` command accepts a stack description in the form of a Compose file.

- Some features are ignored by stack when using a compose file (e.g. build)

This results in an image named `webapp` and tagged `tag`, built from `./dir`.

> **Note when using docker stack deploy**
>
> The `build` option is ignored when deploying a stack in swarm mode The `docker stack` command does not build images before deploying.

- More suited for Production scenarios.

Importance for CI/CD

- Some good features to CI/CD:
  - Update the image used to a newer version
    - *docker service update --image myapp:1.2.1 <servicename>*
  - Adding an environment variable and remove a port
    - *docker service update --env-add NODE_ENV=production --publish-rm 8080*
  - Change number of replicas of two services
    - *docker service scale web=8 api=6*
- For Stack files, we only have to change YAML parameters.

Importance for CI/CD - rolling update

4. Now you can update the container image for `redis`. The swarm manager applies the update to nodes according to the `UpdateConfig` policy:

```
$ docker service update --image redis:3.0.7 redis
redis
```

The scheduler applies rolling updates as follows by default:

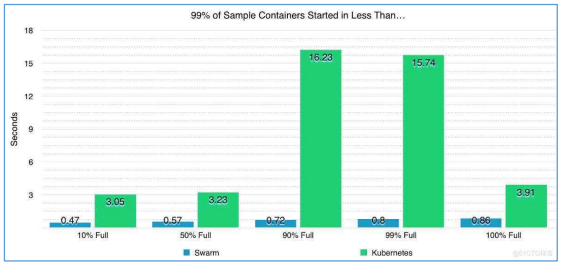
- Stop the first task.
- Schedule update for the stopped task.
- Start the container for the updated task.
- If the update to a task returns `RUNNING`, wait for the specified delay period then start the next task.
- If, at any time during the update, a task returns `FAILED`, pause the update.

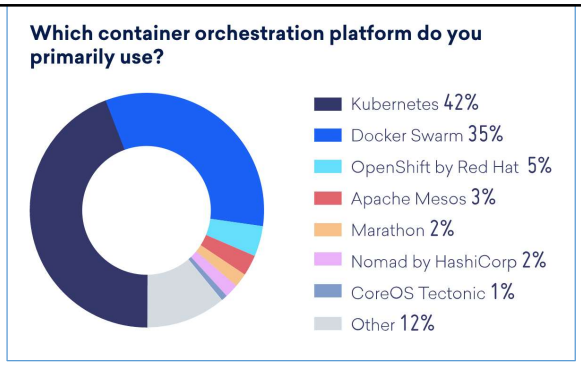
https://docs.docker.com/engine/swarm/swarm-tutorial/rolling-update/

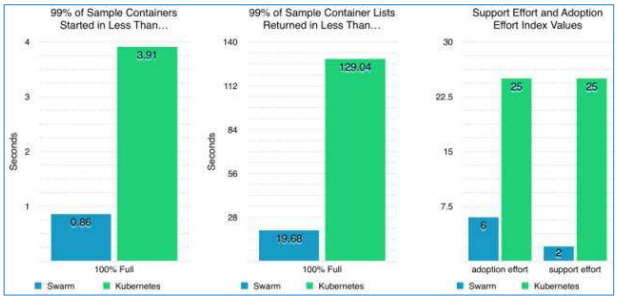
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Swarm deprecated?



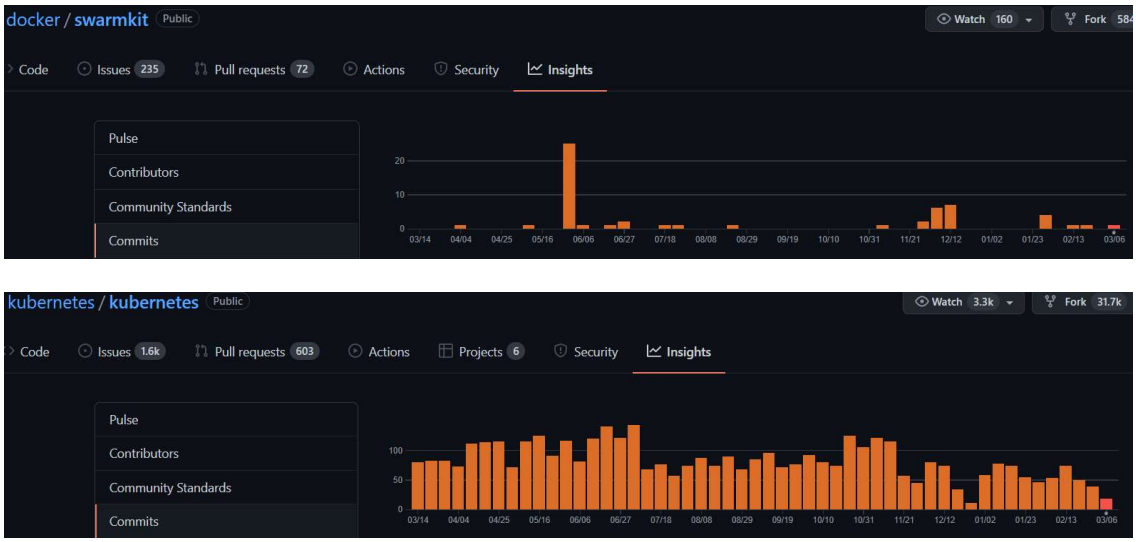






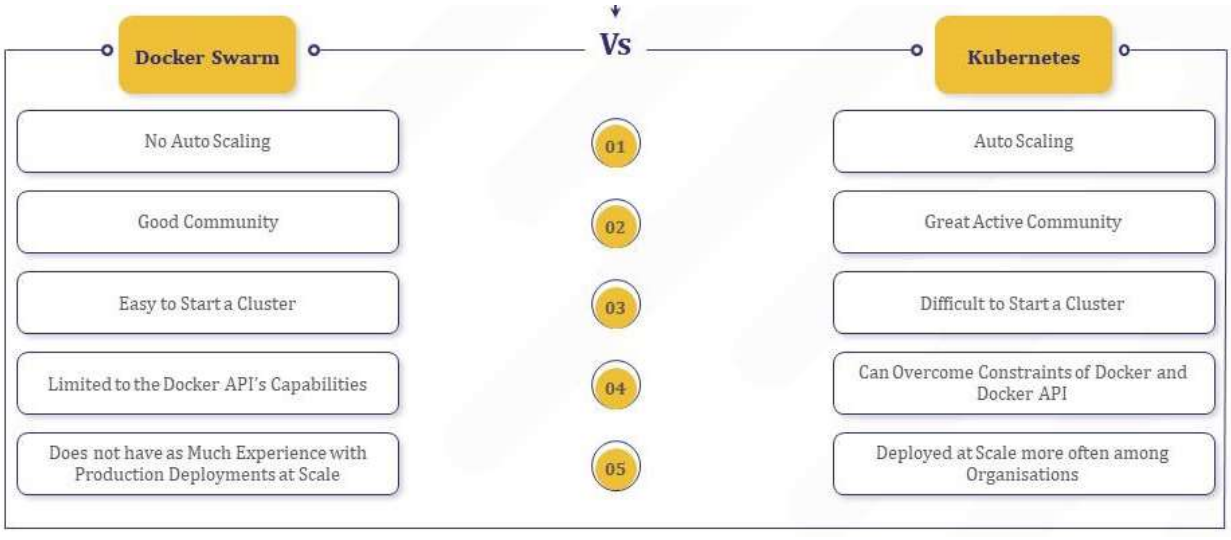
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Swarm deprecated?



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Swarm deprecated?



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Swarm deprecated?

	Swarm	Kubernetes
Controller	Manager	Master
Slave	Worker	Nodes
Workload Definition	Service	Deployment
Deployment Unit	Task	Pod
Scale-out Definition	Replicas	Replica Set
Service Discovery	DNS	DNS
Load Balancing	Ingress	Service
Port	PublishedPort	Endpoint
Storage	Volumes	Persistent Volumes / Claims
Network	Overlay	Flat Networking Space

Some bibliography

- https://docs.docker.com/engine/swarm/key-concepts/
- http://thesecretlivesofdata.com/raft/
- https://docs.docker.com/engine/swarm/
- https://docs.docker.com/engine/swarm/services/
- https://docs.docker.com/engine/swarm/how-swarm-mode-works/services/
- https://www.oreilly.com/library/view/continuous-delivery-with/9781787125230/5f79612d-cebc-4267-b8f5-b3b5b37ed49c.xhtml
- https://docs.docker.com/compose/compose-file/compose-file-v3/
- https://docs.docker.com/engine/swarm/stack-deploy/
- https://javamana.com/2021/06/20210627215228212G.html